

A bio-inspired ferroelectric tunnel FET-based spiking neuron for high-speed energy efficient neuromorphic computing

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ABSTRACT

In this study, a novel spiking neuron based on double-gate ferroelectric tunnel field effect transistor (DG-FE-TFET) is proposed in the 50 nm technology node. Through calibrated simulations in Atlas TCAD, it is verified that by leveraging the forward transfer characteristics, the device accurately emulates the spiking patterns exhibited by biological neurons. The device works on tunneling mechanism which ensures low voltage operation and thereby reduces overall energy consumption. The proposed neuron consumes an energy of 0.58 aJ per spike, which is 1.29, 41.38, 7.75×10^5 , 12.9×10^5 , 1.89×10^5 , 13.79×10^6 , and 7.75×10^7 times lesser than V-DGTFET, FE-JLFET, Z^2 -FET, DG-TFET, FD-MOSFET, Ge-MOSFET, and Si-MOSFET. Notably, the neuron operates without the need for additional reset circuitry and achieves spiking frequency of 344 GHz, which can be attributed to the sharp switching characteristics of the proposed device. This greatly simplifies implementation and facilitates a higher neuron density for energy-efficient neuromorphic chips with high-speed capabilities. Finally, based upon the proposed neuron, a multi-layer spiking neural network (SNN) is designed in Python platform. The network is implemented for signal classification with an accuracy of 84.4%.

1. Introduction

Neuromorphic computing, an emerging field with roots dating back to the 1980s, has gained significant attention and undergone remarkable advancements in recent years due to the rapid progress in artificial intelligence. **It employs artificial neural systems with analog circuitry to mimic the functionality of the human brain.** In the evolution of neural network models, the 1st generation perceptron operates by applying a threshold to the weighted synaptic input to produce binary outputs. The 2nd generation artificial neural networks (ANNs) are characterized by their multi-layered architecture. They are widely used in deep learning for complex tasks like recognizing images and understanding language. The 3rd generation, SNNs, represents a remarkable leap in emulating biological neural networks. SNNs model neurons that communicate through generated spikes of electrical activity. This close resemblance to the human nervous system has captivated researchers in the pursuit of neuromorphic computing.

Within a biological neuron, an essential role is played by the bilayer lipid cell membrane, which acts as a leaky capacitor and is responsible for the integration of incoming spikes. As the membrane potential gradually builds up and reaches a critical threshold, it triggers the firing of the neuron and generates a spike. This process is referred to as "integrate-and-fire," and it is represented by the widely used **leaky integrate-and-fire (LIF) model**. The LIF model is favoured for its simplicity in design, achieved by employing a minimal number of circuit elements. An illustration of biological neuron model, the equivalent circuit of the LIF neuron, and schematic

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of a typical SNN are shown in Fig. 1(a), (b), and (c), respectively.

Traditional artificial spiking neurons are typically constructed using complex complementary metal-oxide semiconductor (CMOS) circuits [1], which involve multiple transistors to create a single neuron. This complexity results in low area and energy efficiency. In response to the need for better efficiency, there has been a push to develop new spiking neurons that rely on a single device. These devices, due to their unique operating principles, can perform the basic functions of spiking neurons with simpler circuits and offer high scalability. However, the main drawback is that many of these spiking neuron devices lack self-reset mechanisms, and hence require the use of additional circuits for initialization, which adds extra area and energy overhead. Therefore, there is ongoing interest in the development of CMOS-compatible compact 1T spiking neurons [2–8]. Another challenge in SNNs is energy consumption. In the context of edge computing in the era of the Internet of Things (IoT), low-power technologies like SOI are crucial. These technologies aim to minimize energy usage while maintaining efficient processing power, making them highly relevant for SNNs.

Concerning this issue, TFETs have gained significant attention in recent times due to their notable characteristics, including subthreshold swing (SS) below 60 mV/decade, low leakage current, and reduced power dissipation compared to conventional MOSFETs. In this work, a novel Indium antimonide (InSb) based DG-FE-TFET is proposed for energy efficient 1T LIF neuron. The device works on tunneling mechanism and does not require additional circuitry for its operation. This greatly simplifies the neuron design, ensures higher integration density, and reduces the overall energy consumption. The rest of the paper is organized as follows. Section 2 explains device description, working principle, and characteristics. Results are discussed in section 3, and the applicability of the proposed neuron in signal classification is explained in section 4. The paper is concluded in section 5.

2. Structural description and model calibration

The architectural details of the proposed DG-FE-TFET are shown in Fig. 2(a). The device is simulated at channel length of 50 nm, and InSb thickness of 20 nm. The source has a p-type doping of $1 \times 10^{20} \text{ cm}^{-3}$ and the drain and channel have n-type doping of $5 \times 10^{19} \text{ cm}^{-3}$ and 1×10^{17} , respectively. The heavily doped 3 nm n+ pocket at source-channel junction effectively narrows the tunneling width and enhances the lateral electric field, resulting in a noticeable improvement in the overall performance of the device. A gate oxide comprising of 4 nm layer of ferroelectric HfZrO₂ (HZO), with a composition ratio of Hf: Zr = 1:1, has been employed. This HZO layer exhibits specific properties, including remanent polarization, $P_r = 10 \mu\text{C}/\text{cm}^2$; coercive field, $E_c = 1.2 \text{ MV}/\text{cm}$; and dielectric constant, $K = 43$. These chosen values have been optimized to attain non-hysteresis behavior [9], which is an important requirement for achieving ultra-low power neuronal operation of the device.

InSb, a group III-V compound semiconductor, demonstrates a compelling set of characteristics, including small effective charge carrier mass (78000 $\text{cm}^2/\text{V}\cdot\text{s}$), low effective bandgap (0.17 eV), enhanced ON current, steeper turn-on behavior, high electron mobility, and efficient auger recombination control [11]. Therefore, it enhances the band-to-band tunneling rate (BTBT) according to equation (1):

$$P(T) \propto \exp \left[-A \left(\frac{E_g^{\frac{3}{2}} \sqrt{m^*}}{E_g + \Delta\phi} \right) \right] \quad (1)$$

where $P(T)$ gives the tunneling probability, as approximated by Wentzel–Kramer–Brillouin approach, m^* is the effective mass of carrier, $\Delta\phi$ is tunneling window i.e., energy difference between maxima of valance band and minima of conduction band minima, and A is a constant.

The device simulations have been performed in Atlas TCAD device simulator [12]. The various models used during the simulations

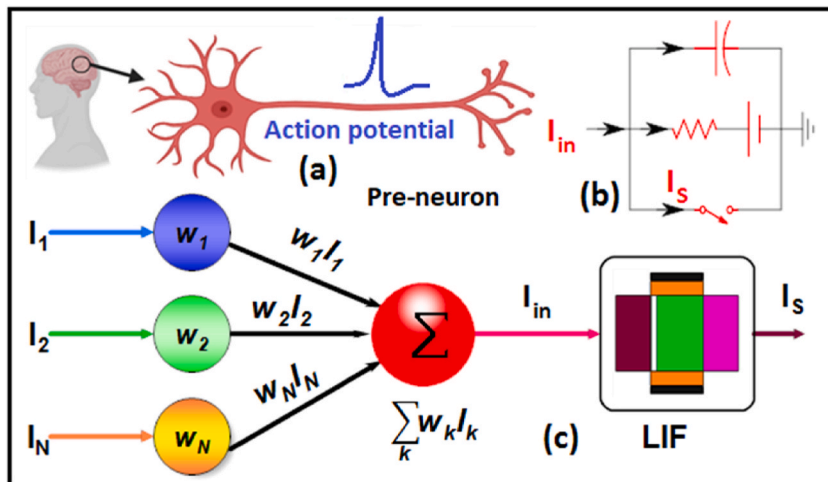


Fig. 1. (a) Representation of biological neuron (b) Equivalent circuit of the LIF neuron (c) Schematics of a typical SNN.

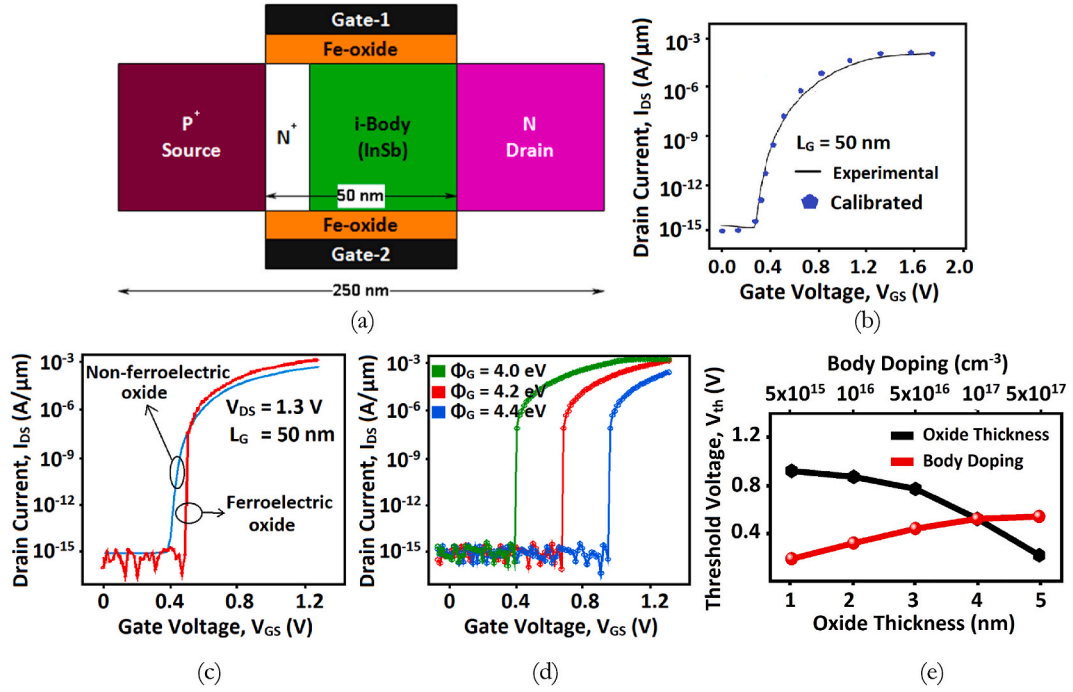


Fig. 2. (a) Schematics of the proposed device (not drawn to scale) (b) TCAD model calibration against the experimental work done by Boucart et al. [10]. (c) Transfer characteristics of the device with ferro and non-ferro gate oxides. Ferroelectric gate oxide shows sharp transition and hence low SS. (d) Effect of gate work function on threshold voltage. (e) Effect of ferroelectric oxide thickness and body doping concentration on threshold voltage.

include *CONSRH*, *DRIFT.DIFF*, *FERMI*, *CONMOB*, *FLDMOB*, *BTBT*, and *FERRO*. To calibrate the TCAD models, the experimental results reported in Ref. [10] have been reproduced, as shown in Fig. 2(b). The simulated results show good agreement with the experimental data. The transfer characteristics of the proposed device are shown in Fig. 2(c). The enhanced ON-current and SS can be attributed to the use of FE-material [9] and *n-pocket* [13]. Theoretically, the SS of a FET is given as follows:

$$SS = \frac{\partial V_G}{\partial(\log_{10} I_D)} = \underbrace{\frac{\partial V_G}{\partial \psi_s}}_x \underbrace{\frac{\partial \psi_s}{\partial(\log_{10} I_D)}}_y \quad (2)$$

where V_G , I_D , and ψ_s represent gate voltage, drain current, and surface potential, respectively. The impact of the ferroelectric material

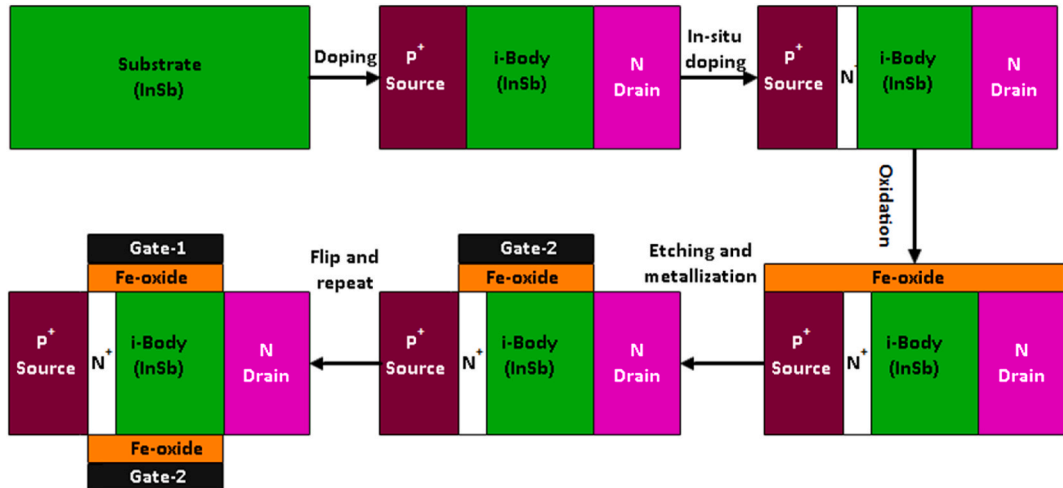


Fig. 3. A possible process flow for fabrication of the proposed device.

reduces the " x " term, while the tunneling process influences the " y " term of equation (2). This results in reduction of SS below theoretical limit of 60 mV/decade for an ideal conventional MOSFET [14]. The sharp switching of the proposed device is responsible for low energy consumption and high spiking frequency of the neuron.

The effect of gate work function (Φ_G), ferroelectric oxide thickness, and body doping concentration on the threshold voltage (V_{th}) of the device has also been studied, as shown in Fig. 2(d). It has been observed that as Φ_G is increased, the breakdown voltage also increases. This is because a higher value of the metal work function raises the flat-band voltage, which in turn increases V_{th} . From Fig. 2(e)—a nonmonotonic dependence of V_{th} on the doping concentration and ferroelectric oxide thickness is observed. These results are in consistent with the previously reported works [15,16].

A possible process flow of the proposed device is shown in Fig. 3. To start with, first a substrate of InSb is doped with donor and acceptor dopants to form drain and source, respectively. A 3 nm pocket is created at the source-channel junction by in-situ n-type doping of $5 \times 10^{19} \text{ cm}^{-3}$. Next, ferroelectric oxide layer is grown by the process of atomic layer deposition. This is followed by metallization of electrodes and etching out of unwanted oxide. Finally, the device is vertically flipped to perform oxidation and metallization of the upper gate [13].

3. Results and discussions

The operational mechanism of the proposed device as LIF neuron can be comprehended through the circuit arrangement presented in Fig. 4(a). The emulation of spiking behavior akin to that of a biological brain is achieved by leveraging the forward transfer characteristics of the device. Initially, the device is assumed to be in the OFF state. Upon the application of a pulsating input current (I_{in}) of 1 μA to the device, the drain capacitor undergoes linear charging, leading to an increase in the output voltage (V_O). This phase is commonly referred to as temporal integration. Subsequently, as the integrated voltage attains the threshold level (V_{th}) of 0.5 V, the device turns ON due to tunneling mechanism, and a current spike is observed, as depicted in Fig. 4(b). As a result, the body capacitor discharges and V_O decreases. When the voltage reaches V_{rest} , the device switches to initial OFF state. The LIF behavior can also be verified from energy band diagrams (EBDs) and BTBT rate observed at different instants. Fig. 4(c) and (d) show the initial behavior of the device with zero tunneling rate and negligible band bending. However, when the device approaches V_{th} , a high tunneling rate is achieved due to significant band overlapping, as shown in Fig. 4(e) and (f).

3.1. Dependence of spiking frequency

A spiking neuron stores information in the frequency of spikes, and not in the profile of a particular spike. To gain insights into the spiking behavior, a pulsating current (I_{in}) with magnitude of 1 μA , duty cycle of 10%, and frequency of 10 GHz is applied. Consequently, the drain capacitor begins to charge, and a spike train is observed. However, when the frequency of input current is reduced to 5 GHz, the charging process slows down, thus decreasing the spiking frequency, as shown in Fig. 5. Additionally, the spiking frequency is a function of drain capacitance and magnitude and duty cycle of input current. A decrease in the capacitance reduces the charging time and increases the spiking rate. The dependence of spiking frequency on the magnitude of input current is shown in Fig. 6(a). It can be observed that when the magnitude is varied from 1 μA to 5 μA , the spiking frequency increases from 10 GHz to 46 GHz. This can be attributed to the fact that a higher magnitude of input current speeds up the charging process and thus increases the spiking frequency. Through extensive simulations in Atlas TCAD, it has been observed that the highest spiking frequency (f_s) that can be achieved by the proposed neuron is 344 GHz at input current of 37 μA . The maximum frequency is limited by the charging-discharging time of the drain

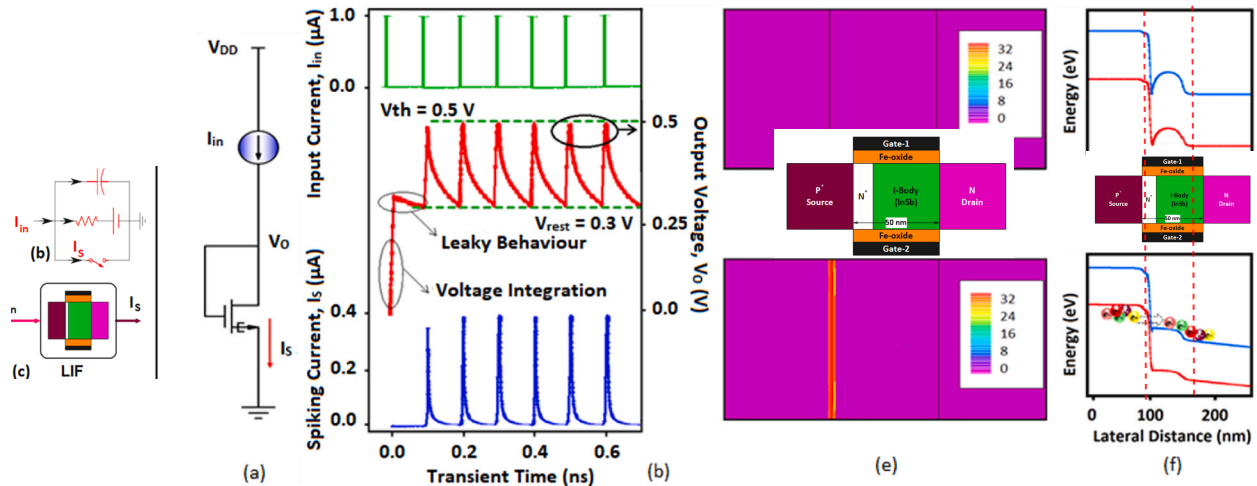


Fig. 4. (a) Biasing arrangement of DG-FE-TFET for LIF neuron. (b) Spiking behavior of the proposed neuron. (c), (d) Tunneling rate and EBD, when device is in the OFF state. (e), (f) Tunneling rate and EBD, when V_O approaches V_{th} , and the device switches to the ON state.

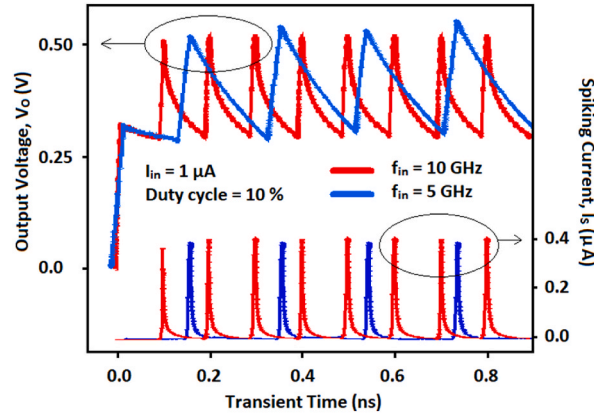


Fig. 5. Dependence of spiking frequency of the proposed neuron on frequency of input current. A higher frequency of input current speeds up the charging process and thus increases the spiking frequency.

capacitor. Since the capacitor discharges through the device, its switching behaviour governs the maximum achievable spiking frequency. While a **higher spiking frequency holds significance in the design of high-speed neuromorphic architectures, it concurrently results in a larger average power consumption.** As previously explained, the **increase in Φ_G results in a corresponding rise in V_{th} .** Consequently, this contributes to an **increase in energy consumption,** as shown in Fig. 6(b).

3.2. Energy and power consumption

The energy per spike, E_{spike} , and average power consumption, P , of a spiking neuron are as follows:

$$E_{spike} = V_{th} \times I_s \times t_{spike} \quad (3)$$

$$P = f_s \times E_{spike} \quad (4)$$

With the help of data obtained from Fig. 4(b), the energy consumption of the proposed neuron is calculated as follows:

$$E_{spike} = 0.5 \times 0.4 \times 10^{-6} \times 2.9 \times 10^{-12} = 0.58 \text{ aJ} \quad (5)$$

And the average power consumption is:

$$P = 344 \times 10^9 \times 0.58 \times 10^{-18} = 0.2 \text{ μW} \quad (6)$$

Notably, the energy and power efficiency of the proposed neuron are significantly improved. This can be attributed to the **lower SS** of the device. **Lower SS ensures that temporal duration of output spike** is very small, causing the device to consume lesser energy. Pertinent to mention that the energy consumption is 1.29 times lower, and the power consumption is halved compared to the recently reported vertically developed double-gate GaSb/Si TFET-based neuron [17].

3.3. Comparison with the state-of-the-art

The proposed neuron is compared with the state-of-the-art neurons that do not use external circuitry, as shown in Table 1. The comparison is made in terms of technology node, energy consumption per spike, spiking frequency, and methodology. It is evident that

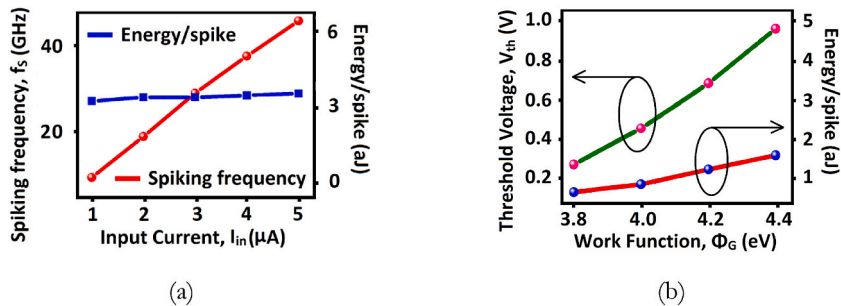


Fig. 6. (a) Spiking frequency and energy consumption per spike as a function of magnitude of input current. (b) Effect of Φ_G on threshold voltage and energy consumption of the neuron.

the proposed neuron consumes the least energy among 1T neurons reported in the literature.

4. Signal classification

Physiological signals offer valuable insights into various aspects of bodily functions and any deviations from their normal patterns can potentially signify underlying health issues. For instance, electrocardiogram (ECG) signals possess the capability to identify arrhythmias, whereas electroencephalogram (EEG) signals can unveil abnormalities indicative of epilepsy during seizure episodes. ECG recordings typically encompass muscle depolarization and repolarization, including P, QRS, and T complexes, as shown in Fig. 7(b). Heartbeats are classified into four distinct categories: normal (N), supraventricular ectopic beat (SVEB), ventricular ectopic beat (VEB), and fusion (F). These classifications adhere to the standards set forth by the Association for the Advancement of Medical Instrumentation (AAMI).

In this section, the proposed neuron has been implemented to classify the heartbeats into one of the four target classes. The heartbeat signals are taken from MIT-BIH database [18]. The spike encoder converts analog ECG signal into positive and negative spike trains, as shown in Fig. 7(a). The positive spike (UP) signifies when the input signal exceeds a threshold, while the negative spike (DOWN) denotes when the signal falls below a threshold. By employing this asynchronous encoding scheme, the accurate timing information of input signal is preserved. Also, the spike trains can accurately reconstruct the original input analog signal due to the inclusion of precise time information [19]. An additional CUE channel is used that continuously fires at the end of each heartbeat to initiate a valid output period.

To verify the signal classification ability, a Python-based equivalent model of the proposed TFET neuron is designed by extracting various electrical parameters that characterize the behavior of the neuron. These parameters include device capacitance (0.2 fF), threshold voltage (0.5 V), resting potential (0.3 V), and OFF-state resistance (8 MΩ). Next, a three-layer SNN is designed consisting of 3 neurons in input layer, which correspond to UP, DOWN, and CUE signals; 100 neurons in hidden layer; and 4 neurons in output layer, which correspond to N, F, SVEB, and VEB classes of heartbeats. The network is implemented in PyTorch and trained used Back-propagation Through Time (BPTT) for 200 epochs. The simulation code and classification methodology are adapted from Ref. [19]. For training and evaluating, datasets of 1650 and 350 samples were used, respectively. The network was successful in classifying heartbeats with the maximum accuracy of 84.4%, as shown in Fig. 7(c). This can also be cross verified by the dark diagonal line of the confusion matrix shown in Fig. 7(d).

5. Conclusion

This paper introduces a novel spiking neuron design based on a double-gate ferroelectric tunnel field effect transistor in the 50 nm technology node. Through calibrated simulations, it is validated that the neuron accurately replicates biological neuron spiking patterns with low energy consumption. The design does not need extra circuitry and spikes in GHz range, simplifying the design and allowing for higher neuron density, which is crucial for large-scale neuromorphic chips. Unlike impact ionization-based neurons, the DG-TFET uses tunneling mechanisms, reducing energy and power consumption. Finally, an SNN has been designed for ECG classification with an accuracy of 84.4%. This work advances energy-efficient neuromorphic computing, holding potential for artificial intelligence and computational neuroscience.

Ethics approval

Not applicable.

Consent to participate

Not applicable.

Table 1

Benchmark with the state of the art.

Ref.	Device Type	Tech Node	Spiking Frequency	Energy/spike (J)	Methodology
[2]	Si-MOSFET	880 nm	0.8 KHz	4.5×10^{-11}	Experimental
[3]	Ge-MOSFET	800 nm	1 KHz	8×10^{-12}	Simulation
[4]	MOSFET	400 nm	1 KHz	3.8×10^{-12}	Simulation
[5]	FD-MOSFET	100 nm	1 MHz	1.1×10^{-13}	Simulation
[6]	DG-TFET	50 nm	12 KHz	7.5×10^{-13}	Simulation
[7]	Z ² -FET	3 μm	180 Hz	4.5×10^{-13}	Experimental
[8]	FE-JLFET	20 nm	1 GHz	24×10^{-18}	Simulation
[17]	V-DGTFET	20 nm	534 GHz	0.75×10^{-18}	Simulation
This work	DG-FE-TFET	50 nm	344 GHz	0.58×10^{-18}	Simulation

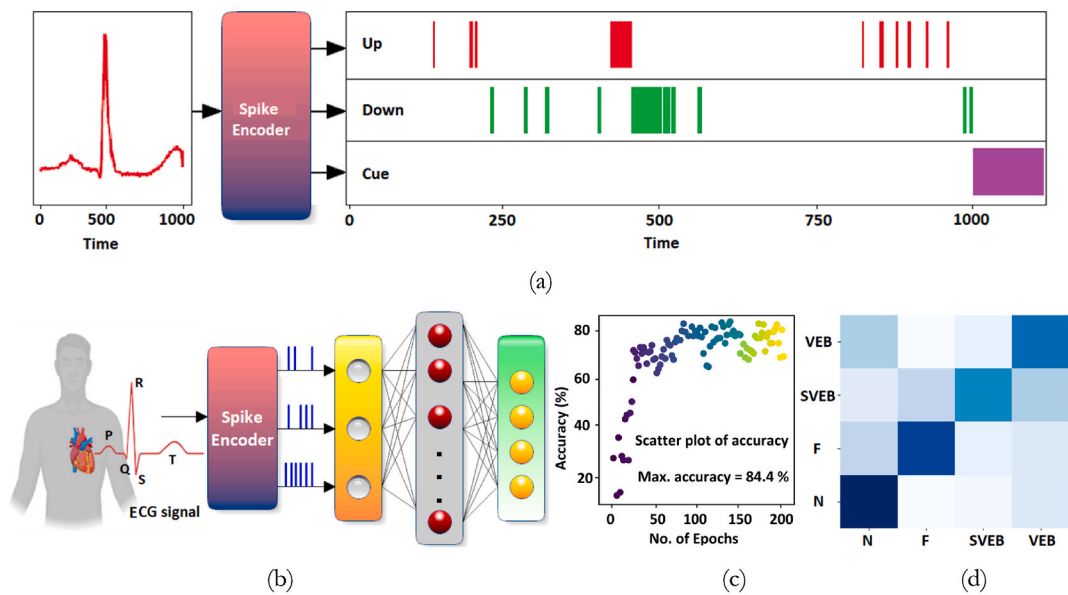


Fig. 7. (a) The spike encoder converts ECG heartbeat into UP and DOWN spike trains, followed by CUE signal indicating a valid output period. (b) Schematics of a typical ECG and the proposed SNN. (c) Scatter plot of accuracy, showing the maximum accuracy of 84.4%. (d) The confusion matrix to validate the functionality of the SNN for the 4-class (N, F, SVEB, VEB) heartbeat classification.

Consent for publication

Yes.

Availability of data and materials

Not applicable.

Competing interests

Not applicable.

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Authors' contributions

All authors have contributed equally.

Disclosure of potential conflicts of interest

There is no conflict of interest among the authors while submitting the manuscript.

Research involving Human Participants and/or Animals

Not applicable.

Informed consent

All authors have been informed before submitting the manuscript.

CRediT authorship contribution statement

Mudasir A. Khanday: Writing – original draft, Validation, Software, Investigation, Formal analysis, Conceptualization. **Farooq A. Khanday:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Conceptualization.

Declaration of competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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